

aiSTROM starter kit

LLM and agents edition | Practical guide, templates and worked examples

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This PDF is the reading edition. Editable Markdown files and the separate fillable canvas are supplied in the starter-kit download. The examples are fictional. Version 0.2 is a proposed practical extension of the 2021 aiSTROM framework.

aiSTROM: visual AI strategy toolkit

One overview canvas. Seven dimension sheets. A practical plan for your AI project.

Use aiSTROM for predictive models, computer vision, recommendations, generative AI, LLMs and agents. Start with the opportunity and choose the technology that fits. The accompanying article uses agents as an example; the toolkit covers AI implementation generally.

[Project website](#) · [Original paper](#)

Download

- **Visual toolkit: eight fillable sheets** — overview plus one sheet for each dimension.
- **Overview canvas only** — for a first conversation or a summary of your decisions.

The visual toolkit is the main working tool. No long workbook is required before you start.

Why use it?

A working model or convincing demo leaves implementation choices unresolved. aiSTROM helps a team discover missing data, skills and ownership, compare delivery options, agree what value means and plan how the system will fit into work. It also makes reusable data and capabilities visible, so a project can contribute to future opportunities.

How the sheets fit together

Sheet	Focus
Overview	Frame an opportunity, capture conclusions across seven dimensions, and record the next decision.
1. Data	Sources and future value; collection/acquisition; quality; privacy; storage; lifecycle and reuse.
2. The AI team	Domain expertise; technical skills; communication/design; obtaining expertise; retention; capacity.
3. Organising AI development	Team positioning; authority; portfolio; existing services; build/partner/acquire; agile delivery.
4. Technologies	Approach and baseline; models/adaptation; explainability; human involvement; hosting; integration.
5. KPIs and value	Strategic outcomes; customer value; processes; financial value; model performance; evidence and review.
6. Risk level and benefits	Performance uncertainty; bias/ethics/safety; security; dependencies; benefits/risk appetite; SWOT.
7. Culture and education	AI literacy; roles; participation; shared expertise; continuous education; adoption in practice.

The original paper supplies the seven dimensions and many of their underlying topics. The six cards on each sheet are practical groupings, not a claim that the paper used these exact headings. Cards identify their source section or mark an added prompt. Conditional prompts at the bottom address particular system types without making every project an agent project.

Fill in a card

Each dimension has six cards. Read the short prompt and record:

1. **Status:** + Supported, - Gap, ? Unknown, or N/A.
2. **Decision:** use the specific field label to record the proposed choice, measure or arrangement.
3. **+ Strength:** a capability, advantage or supporting evidence you can build on.
4. **- Gap:** a missing capability, constraint or unresolved concern.
5. **Next action:** a concrete step, responsible person and date.

A card may have both strengths and gaps. Choose the status that best describes readiness for the next decision; keep both sides visible. Use ? Unknown when evidence is missing, and explain N/A. The signs are discussion aids: do not add them into a score or let several strengths cancel an unresolved prerequisite.

For example, on **Data** → **Collect, acquire or label**:

- **Decision:** use internal maintenance records and have a technician label failure causes.
- **+ Strength:** three years of linked asset and repair records.
- **- Gap:** inconsistent failure codes; annotation effort is unknown.
- **Status:** ? Unknown.
- **Next action:** maintenance lead samples 50 records and estimates labelling effort by the agreed date.

This is a fictional illustration, not a reported deployment.

Suggested working session

1. **Frame the opportunity.** Name the problem, affected people, desired outcome, scope and simpler alternatives in the overview's first box.
2. **Scan the seven dimensions.** Bring together business, domain and technical perspectives. Start with the sheets that could most change your decision.
3. **Investigate the gaps.** The first conversation will surface questions; collecting evidence can happen afterwards. Use short entries and link existing records rather than writing essays.
4. **Connect the dimensions.** A choice to fine-tune affects data, skills, hosting and costs. A new workflow affects roles, education and KPIs. Record dependencies at the bottom of each sheet.
5. **Return to the overview.** Capture the main conclusions and decide whether to explore, pilot, revise, defer or stop. Name the next evidence, owner and review date.

Use one set per opportunity. Compare multiple overview canvases when making portfolio decisions. Revisit affected sheets when goals, data, people, models, providers or operating conditions change.

Using the files

All eight sheets are interactive PDF forms, designed at **A3 landscape** for a workshop table or screen. Print at A3 for comfortable handwriting. A4 printing reduces both text and writing space.

Use a PDF reader that supports forms; save and reopen your copy to check that entries were retained. Keep entries concise and put longer evidence in linked project records. The same overview is available separately, so there is no need to print the entire set.

Feedback

What did the sheets help you notice? Which decision changed? Where did you need more space or a clearer prompt? Share feedback through the [project website](#), removing confidential information.

Reference

Herremans, D. (2021). **aiSTROM—A Roadmap for Developing a Successful AI Strategy**. *IEEE Access*, 9, 155826–155838. [Published paper](#) · [Open author version](#).

Dorien Herremans · Visual AI strategy toolkit · v0.3

Optional deeper references

The existing implementation workbook, specialist sheets and decision records remain available when a topic needs more investigation. They are supplementary material, not the required route through the visual toolkit.

aiSTROM project canvas

LLM and agents edition · v0.2 · Dorien Herremans

[Original paper](#) · Guide

This canvas summarises the detailed workbook; it does not replace the analysis. Copy this file for each opportunity. Use short answers and link to supporting evidence. Prefix uncertain statements with **Assumption** or **Unknown**; give unknowns an owner and due date. Completion of the form is not deployment approval.

Project / workflow:

Team / organisation:

Canvas owner:

Date / revision:

Current stage: Explore / scoped pilot / operational review

Opportunity: what work should improve?

- People and task:
- Current problem and evidence:
- Desired outcome and scope:
- Current process / simpler alternative:
- Why examine this opportunity now?

1. Data: what information can it use?

- Approved sources, owners and freshness:
- User and system access boundaries:
- Missing or conflicting information response:
- Provider use of prompts, retrieved data, tool results and outputs; training, retention, access and deletion:
- Output / log storage and retention owner:

2. Team: who owns delivery and quality?

- Workflow owner:
- Technical owner:
- Quality / evaluation owner:
- Reviewers, support and missing expertise:

3. Organisation: how will it operate?

- Buy / configure / build; reason:
- Budget and decision owner:
- Cross-team dependencies:
- Ongoing support and exit approach:

4. Technology and autonomy: what may it do?

- Provisional approach and simpler option considered:
- Model source / adaptation and hosting choice; provider and product:
- Performance, explainability, cost and exit evidence:
- Allowed actions and enforced limits:
- Actions requiring approval; approver:
- Prohibited actions; escalation route:

5. Value and evaluation: what would justify continuing?

- Baseline and how it will be measured:
- Target outcome, denominator and time window:
- Quality / failure measures and thresholds:
- Operating cost, review effort and maintenance:

6. Risk and recovery: what could go wrong?

- Benefits and SWOT; acceptable risk level:
- Consequential errors, bias and accountability:
- Prompt injection, tool compromise, access misuse and isolation failure:
- Prevention and detection:
- Stop conditions and responsible person:
- Recovery / rollback / fallback; limitations:

7. Adoption and learning: how will people use it?

- Pilot users and task-specific training:
- Feedback / corrections route:
- Existing workflow fallback:
- Review cadence and owner:

Decision and next evidence

Decision: Explore / pilot / revise / defer / stop

Reason:

Authorised scope now:

Unresolved prerequisites:

Next evidence / action:

Action owner and due date:

Decision-maker and review date:

Use the detailed workbook and specialist sheets to develop the decisions summarised here.

Worked example: sourcing replacement components

Fictional teaching scenario · all quantities and thresholds are illustrative design choices. No deployment or measured improvement is claimed.

aiSTROM v0.2 · [Toolkit](#) · Completed canvas

Opportunity and alternatives

A manufacturer faces occasional delays from component suppliers. Buyers manually compare alternative quotations with engineering requirements. The initial idea is an agent that identifies alternatives and places orders.

The team compares better stock alerts, a curated supplier list, quote-comparison assistance and automatic purchasing. It chooses to investigate comparison and draft preparation first. Better purchasing decisions are the goal; autonomous ordering is not a prerequisite.

Scope: compare quotes from already approved suppliers for non-safety-critical components, using the current approved specification. Provide an evidence-linked comparison and a draft order for a buyer. Engineering substitutions, supplier onboarding, external communication, live order creation, payment and bank-detail changes are excluded.

1. Data strategy

Required information includes specifications, approved suppliers, quotations and relevant inventory. The engineering owner resolves component revisions; procurement owns quote validity and terms; IT owns access enforcement. Quotes require explicit units, quantities, currency, validity, delivery terms and certification evidence where applicable.

For initial offline testing, use synthetic or appropriately de-identified records. Before live data are allowed, complete S1 for the actual product, account and deployment, including confidential quote content, retrieved specifications, outputs, logs and feedback. Record provider training, retention and access terms separately. An internal application using a hosted model is still an external data flow.

Unclear units, missing certification or conflicting specifications must be flagged for a person. The system must not silently infer equivalence between components.

2. Team

Procurement lead: business outcome and scope. Buyer: workflow and usability. Engineer: specification judgement. Integration owner: tools and access. Security/data owner: deployment and boundaries. Evaluation owner: rubric, independent review and evidence. Assign backup coverage and time for support.

Training or external expertise may fill gaps. The team must retain enough domain and technical knowledge to review a vendor's work and operate the process afterwards.

3. Organisation

Compare configuration of existing purchasing software with an internally developed comparison service and an external developer. The business unit owns outcomes; IT supplies shared integration/security expertise. No delivery option is selected before checking the requirements.

The procurement lead authorises the offline experiment. Live deployment requires the data owner and integration/security owners to confirm their conditions. The existing buyer process remains the fallback. Record how to export evaluation material and disable or replace the service.

4. Technologies and models

Compare a structured extraction/template baseline, a hosted commercial model with retrieval, and an existing open-weight model operated in an approved environment. Fine-tuning is considered only if observed errors justify it and suitable data exist. Training from scratch is not justified by evidence at this stage.

Evaluate task quality, provider data terms, operating skills, cost, latency, model changes and switching effort separately. No model or provider is approved in advance.

Initial authority:

- Read only the approved test collection.
- Extract quote fields and produce a private comparison with source references.
- Prepare a draft order outside the live purchasing system.
- No tools for messaging suppliers, writing live orders, changing vendor details or paying.
- No new tool installation, delegation or broader credentials.

Missing evidence causes escalation. A later proposal to create orders would need new permissions, transaction controls, approval requirements and evaluation. It is not covered by this decision.

5. KPIs and evaluation

The first experiment asks whether assistance creates accurate, reviewable comparisons and reduces buyers' total effort compared with the existing template. It does not test purchasing autonomy or prove fewer stockouts.

Illustrative test design: 30 held-out task packages covering routine quotes, unit/currency differences, expired offers, missing fields and conflicting specifications. Use separate development material. Have both approaches process the same packages; counterbalance reviewers' condition order and record possible familiarity effects. An engineer and buyer adjudicate the reference judgements. Repeat ten packages three times to inspect variability rather than assuming one successful run is representative.

Measure critical field accuracy, surfaced constraints, unsupported claims, escalations, review/rework time and tool activity. Define the field denominator from the final annotated test set; report counts alongside percentages.

Proposed local progression conditions, to be confirmed by the responsible team before running the test:

- All specification conflicts and expired-offer cases in the fixed test set are flagged.
- No unsupported component-equivalence claim and no prohibited tool action is observed.
- At least 95% of annotated critical fields are extracted correctly, with every critical error reviewed by type.
- Median total buyer time is below the comparator, without worse constraint-detection results.
- S1 conditions and independent access tests pass before any live confidential data are used.

These are exploratory thresholds for this scenario, not validated aiSTROM standards. A small sample with no observed severe failures cannot establish safety. Record sensitivity to task mix and reviewers. Passing supports consideration of a bounded draft-only live pilot, not automatic purchasing.

The cost record includes implementation, model or hosting cost, review, evaluation, content maintenance and support. No ROI or savings claim is made before collecting those measurements. Delivery reliability and stockouts need a longer observation window.

6. Risk, benefits and recovery

Potential benefits: faster sourcing, clearer comparisons and more buyer time for negotiation. Weaknesses: inconsistent quote formats and incomplete specifications. Opportunities: reuse of better supplier data. Threats: changing terms, dependency on a provider and malicious or misleading external documents. Each SWOT item leads to an action or explicit uncertainty.

Test a supplier document containing instructions that conflict with the task, using synthetic material. Confirm the assistant cannot send documents to external destinations or access vendor-write/payment tools. Assess tool provenance, credentials and network isolation independently; a well-written prompt is not a security boundary.

Also test ordinary safety failures: confused units, unsuitable substitutions and repeated draft generation. Log sources and tool attempts. Suspected disclosure, prohibited access or a missed critical constraint pauses the experiment. The integration owner can revoke access; procurement returns to the manual process; security handles incident investigation. Disclosure may not be reversible.

Commercial accountability includes checking whether draft language could imply an accepted offer. Staff must treat external communication as a separately authorised action.

7. Adoption and education

Buyers and engineers practise with ambiguous and suspicious examples. They inspect evidence, report corrections and can discard outputs. Weekly review examines the source of errors and actual checking effort. Maintain procurement and engineering expertise; do not reward accepting drafts merely to increase usage.

Recorded decision

Explore through an offline comparison experiment. Confirm test-data permission, annotate the evaluation set and verify the draft-only boundary. The procurement lead owns the decision after reviewing quality, time, cost and security evidence. A live pilot remains conditional on S1/S2 and further approval within the organisation. Autonomous ordering remains outside scope.

What aiSTROM changes in this fictional example: a broad automation idea becomes a portfolio choice, a model/deployment assessment, a team and operating arrangement, and a specific experiment. It does not predetermine the model, provider, or decision to deploy.